

Overall Brain Signals ADHD training

STEP 1: Data Collection

- EEG dataset (ADHD + Non-ADHD)
 - Each subject:
 - 19 EEG channels
 - ~ 14,304 samples per channel
 - Sampling rate = 128 Hz
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◆ STEP 2: Load EEG Data (.mat files)

- Read MATLAB .mat files
- Extract EEG matrix
- Transpose to shape:

(19 channels × time samples)

◆ STEP 3: Temporal Windowing

You defined:

WINDOW_SIZE = 256 → 2 seconds

STEP_SIZE = 128 → 50% overlap

Then:

- Split long EEG into small 2-second segments
- Each window becomes one training sample

Example:

(19, 14304) → many (19, 256) windows

This increases dataset size and captures temporal dynamics.

◆ STEP 4: Label Assignment

- ADHD → label 0
- Control → label 1

Each window inherits subject label.

◆ STEP 5: Combine All Subjects

- Concatenate ADHD_part1
- ADHD_part2
- Control_part1
- Control_part2

Final dataset shape:

(samples, 19, 256)

◆ STEP 6: Normalization (Standard Scaler)

For each window:

$(\text{value} - \text{mean}) / \text{standard deviation}$

- Stabilizes training
 - Prevents dominance of large values
 - Improves convergence
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◆ STEP 7: Reshape for CNN

Convert from:

(samples, 19, 256)

To:

(samples, 19, 256, 1)

Why?

Because Conv2D expects 4D input.

◆ STEP 8: Train–Test Split

80% → training

20% → testing

With stratification to maintain class balance.

Purpose:

- Evaluate generalization
 - Avoid overfitting
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◆ STEP 9: Build CNN Model

Architecture:

Conv2D → MaxPooling

Conv2D → MaxPooling

Flatten

Dense

Dropout

Output (Sigmoid)

What CNN does:

- Learns spatial patterns (across electrodes)
 - Learns temporal patterns (across time)
 - Combines them for classification
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◆ STEP 10: Train Model

Using:

- Adam optimizer (Adam intelligently adjusts learning speed for each weight during training)
- Binary crossentropy loss
- 10 epochs
- Batch size 32

Model learns via backpropagation.

◆ STEP 11: Evaluate Model

- Test accuracy
- Confusion matrix
- Precision, Recall, F1-score

Result:

~99% accuracy (window-level)

◆ **STEP 12: Interpret Model (Grad-CAM)**

- Visualized important EEG regions
- Identified temporal areas influencing prediction
- Addressed black-box problem

At glance :

One subject → 110 pieces

All subjects → 16748 pieces

80% train → 13398

20% test → 3350

Train divided in batches of 32

335 batches per round

20 rounds

6700 learning updates

Calculation part :

Data : 19 channels

14,304 time samples

Sampling rate = 128 Hz

Training and testing split :

Training samples = 0.8×16748

= 13398.4 $\approx$ **13398 samples**

Testing samples = 0.2×16748

= 3349.6 $\approx$ **3350 samples**

Windowing :

WINDOW_SIZE = 256

STEP_SIZE = 128

Number of windows = $14304 - 256 = 14048$

$$= \frac{14048}{128}$$

= 109.75 $\approx$ 110 windows for each subject

Window shape = (19,256)

Applying Normalization : for each value

$$z = x - \text{mean} / \text{std}$$

Reshaping for CNN : Original window shape : (16748 , 19, 256)

(samples, channels, time)

Reshape window shape :

(batch_size, height, width, channel)

(16748, 19, 256, 1)

Validation Split :

Real training data : 13398

Validation = 20% of training data = $0.2 \times 13398 = 2679$

Training = $13398 - 2679 = 10719$

Test = 3350

Training data in batches :

Real training samples = 10719

$$\frac{10719}{32} = 334.96 \sim 335$$

Batches per epoch

As total epochs = 20

Total updates: $335 \times 20 = 6700$

Model updates weights **6700 times**